

LAERDAL IV, Norway

WMO: 013550

Lat: 61.103N

Lon: 7.503E

Elev: 16

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.6	13.3	0.3	5.6	19.5	3.3	6.5	17.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	77.7	60.9	73.9	59.2	70.9	58.0	63.5	73.0	61.8	69.9	60.4	67.8	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	77.9	65.4	58.6	73.5	64.0	57.1	69.7	62.3	28.8	72.1	27.7	70.0	26.7	68.2	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	9.1	83.6	4.4	6.2	5.9	88.1	3.3	91.8	0.8	95.3	-2.4	99.8		
(5)			8.4	66.0	4.0	1.6	5.6	67.1	3.3	68.1	1.1	69.0	-1.8	70.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	30.4	30.8	37.0	44.0	51.2	57.1	62.7	59.1	52.7	42.8	36.4	32.0
	DBStd	12.81	7.83	8.43	6.21	5.17	5.36	4.05	5.25	4.01	5.00	6.89	7.39	8.57
	HDD50	3026	608	538	402	191	49	3	0	0	28	239	411	557
	HDD65	7429	1073	958	867	630	429	238	108	188	369	688	859	1022
	CDD50	1118	0	0	0	11	86	216	394	283	110	16	2	0
	CDD65	44	0	0	0	0	0	1	37	6	0	0	0	0
	CDH74	331	0	0	0	0	5	20	277	29	0	0	0	0
	CDH80	58	0	0	0	0	0	0	58	0	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	27.1	2.7	2.3	2.3	1.2	1.3	1.9	2.2	2.4	2.4	2.5	2.5	2.8
	PrecMax	39.7	7.9	6.1	10.3	2.7	2.4	5.2	4.2	4.2	4.0	6.7	6.7	7.0
	PrecMin	18.3	0.0	0.0	0.2	0.4	0.3	0.5	0.7	0.6	0.8	0.7	0.0	0.3
	PrecStd	4.9	2.2	1.7	2.0	0.7	0.6	1.2	0.8	1.0	0.9	1.4	1.8	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.3	51.3	54.3	64.8	74.1	76.5	85.7	77.4	68.8	65.4	57.4	51.8
		MCWB	40.0	41.4	44.8	49.7	56.6	57.3	63.9	58.2	55.0	52.9	46.9	42.0
	2%	DB	46.3	47.9	50.3	60.8	70.0	72.5	80.6	73.4	65.5	60.3	53.2	47.8
		MCWB	38.4	38.8	41.8	48.2	54.9	55.3	63.2	58.7	54.5	50.3	44.1	39.7
	5%	DB	41.8	44.9	48.0	57.0	65.9	69.4	76.7	70.3	63.0	55.0	49.5	44.4
		MCWB	36.1	36.9	40.0	45.7	53.0	54.6	61.6	58.3	54.2	48.1	42.4	37.5
	10%	DB	39.4	42.3	45.7	53.3	61.8	66.6	73.1	67.2	60.8	51.9	45.7	42.4
		MCWB	34.4	35.5	39.0	43.8	50.9	53.8	60.3	57.5	53.0	45.8	41.6	37.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.6	43.0	46.0	51.6	59.1	60.7	66.6	63.5	59.6	54.6	48.5	43.7
		MCDB	45.3	47.9	51.8	62.3	71.0	70.8	79.5	70.6	62.9	59.9	54.2	48.4
	2%	WB	39.6	40.1	43.7	49.0	56.2	58.3	64.6	61.7	57.4	52.1	46.4	41.4
		MCDB	43.7	45.0	48.2	58.9	66.6	66.7	76.1	67.9	61.3	59.2	52.0	45.4
	5%	WB	37.9	38.1	41.7	46.9	54.2	56.8	63.2	60.4	55.9	49.6	44.0	39.7
		MCDB	41.9	43.5	46.0	55.1	63.4	64.1	72.6	66.3	60.1	54.1	47.7	43.9
	10%	WB	35.6	36.5	39.9	45.0	52.3	55.6	61.8	59.1	54.6	47.2	41.8	38.0
		MCDB	38.2	41.4	44.4	51.6	60.5	62.6	70.4	64.8	59.1	50.9	45.7	41.6
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.8	11.3	14.8	16.6	16.5	15.9	13.8	12.1	9.9	8.7	8.0
		MCDBR	10.7	11.1	12.9	20.4	24.1	22.6	23.2	18.7	14.5	12.8	13.1	11.8
		MCWBR	6.2	6.1	7.4	10.4	11.0	9.7	8.4	8.1	6.8	6.7	7.1	7.1
	5% WB	MCDBR	9.5	10.5	12.2	18.4	21.9	17.5	19.9	15.7	13.0	11.1	12.6	11.9
		MCWBR	6.2	6.4	7.7	9.8	10.4	8.5	8.2	7.6	7.0	6.6	7.4	7.9
Clear-Sky Solar Irradiance	taub	0.234	0.280	0.318	0.354	0.360	0.357	0.373	0.364	0.330	0.305	0.244	0.203	
	taud	2.083	2.189	2.137	2.105	2.200	2.354	2.379	2.398	2.472	2.367	2.119	1.976	
	Ebn at Noon	158	221	250	263	272	276	268	260	251	213	149	111	
	Edh at Noon	12	22	32	40	40	35	33	30	23	18	12	8	
All-Sky Solar Radiation	RadAvg	71	232	620	1179	1519	1535	1461	1123	701	322	102	46	
	RadStd	17	67	127	200	234	170	167	110	87	48	17	6	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page